March 25, 1959

MEMO TO: Mr. D. W. Pendery

SUBJECT: Expanded Character Set Printing

1. The major requirement for a speedy evolution of expanded character sets has been in the area of advanced programming system languages. This requirement has grown from the desire to simplify the quantity of information required for program input into computing systems. At the same time, expanded character sets, permit consideration of more complex relationships.

2. The above point is stressed because coincident with the requirement mentioned above, a somewhat less vocal opinion has evolved regarding the use of an on-line line printer for printing normally done on a typewriter.

3. A serious question evolves regarding many approaches to producing the printing associated with the expanded character sets. Where low volume usage is predicted then the total needs of expanded character set users can be satisfied by the Reading and Punching Typewriter. This device used with on-line card readers and card punches permits a reasonable quantity of information to be handled.

   Assuming printed output of approximately 500 to 600 lines we have a printed listing in 30 to 60 minutes. This discussion assumes a low frequency of printing per RPT.

4. Furthermore, for those applications where scientific reports, or similar types of document, require updating or minor revisions the approach taken in paragraph 3 permits a simple manual file maintenance on a deck of cards followed by a typing run on the RPT for those master pages which have been altered. Supplemental information such as large quantities of numerical tabular data, could be produced on system printers at much greater speeds.

5. Where larger volumes of printing, or increased frequency of usage indicate, the choice of an on-line expanded character set printer is felt to be necessary as indicated in the meeting reports of Mr. E.G. Law's. Such a printer has the advantage of making on-line card input directly reproducable at increased printing speeds and with no intermediate operator handling.

A. D. Kulsby

6. The chain printer being developed for Project 7000 usage is sufficiently advanced to permit consideration. In its standard form the chain printer permits printing of character sets whose coding is the same as used with current peripheral equipment. In its modified form, a 48 character subset of a 119 character set is usable where the interval coding is different from currently used codes.

7. Attached is a chart, Figure 1 showing the result of using a Standard (Std.) chain or a "B" chain. Both character set representations appear to be useable on the same printer thus providing for a fully compatible printer, in terms of 48 character sets.

8. As indicated in Mr. Law's reports a realistic appraisal of the value of expanded character sets could be obtained by field testing a limited number of RPT units. Coincidently the Project 7000 Chain Printer can have an expanded character chain developed. Such a set need not necessarily be compatible with the expanded set desired by Los Alamos. The present proposal for a 119 character set for the chain printer has been defined with the Inquiry Station code and character arrangement as a subset.

9. Use of these devices by responsible parties in Applied Programming in conjunction with advanced programming systems development should illustrate the basic requirements for operating systems. Coincidently field testing in selected customer installations would provide data on useful techniques in report generation. Such an experiment would provide indications of the relative desirability of upper and lower case character printing. Note that in any field testing operation it might be simplest to use a printer attached to a 704 or 709 where the word might be considered as 5 - 7 bit characters. rather than 6 - 6 bit characters. Such attachment would require a DSU Adapter or off-line "odd parity" tapes for which a 6 to 7 byte converter could be used.

10. Notes for Figure 1 attached.
    (a) "Std." refers to the character set chain arrangement normally available with the chain printer. The type arrangement is such that normal 705 coding will result in printing the usual character.

(b) "B" refers to the 48 character subset of the 119 character set proposed by Dr. W. Buchholz in his memo to Mr. R. W. Bemer dated February 16, 1959. In this arrangement the alphabetic letter "Oh" is to be printed as O and the digit zero as 0.

E. W. Coffin

E. W. Coffin
Product Planning Representative
Project 7000

EWC:jcj

Attachment
cc: Mr. D. W. Sweeney
Mr. E. G. Law

| | 0 0 | | 0 1 | | 1 0 | | 1 1 | |
|---|---|---|---|---|---|---|---|---|
| | Std. | B | Std. | B | Std. | B | Std. | B |
| 0 0 0 0 | | Blank | - | C | | ⊙ | | 0 |
| 0 0 0 1 | ⊥ | ' | / | D | J | P | A | 1 |
| 0 0 1 0 | 2 | $ | $ | E | K | Q | B | 2 |
| 0 0 1 1 | 3 | / | T | F | L | R | C | 3 |
| 0 1 0 0 | 4 | & | U | G | M | S | D | 4 |
| 0 1 0 1 | 5 | % | V | H | N | T | E | 5 |
| 0 1 1 0 | 6 | ) | W | I | O | U | F | 6 |
| 0 1 1 1 | 7 | * | X | J | P | V | G | 7 |
| 1 0 0 0 | 8 | — | Y | K | Q | W | H | 8 |
| 1 0 0 1 | 9 | . | Z | L | R | X | I | 9 |
| 1 0 1 0 | ∅ | A | + | M | — | Y | ■ | # |
| 1 0 1 1 | # | B | , | N | $ | Z | . | @ |
| 1 1 0 0 | @ | | % | | * | | ¤ | |

Fig. 1

EWC 3/23/59